

Hardware-Validated Peer-to-Peer Energy Trading on Ethereum: DepositLock for Trade Non-Fulfillment Risk and On-Chain Performance Characterization

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ABSTRACT Peer-to-peer (P2P) energy trading enables prosumers to sell surplus electricity directly to consumers without a central intermediary. Blockchain-based smart contracts are a natural fit for this setting because they provide tamper-resistant, programmable settlement. Existing implementations, however, address the problem primarily through simulation: the risk that a trading party may fail to fulfil its contractual obligation—leaving the counterparty uncompensated—has not been treated by prior hardware-validated work. This paper makes three contributions. First, we design and implement *DepositLock*, a Solidity smart-contract mechanism that requires a consumer to lock a collateral deposit equal to the agreed price at the time of order submission; the deposit is released to the prosumer only on confirmed trade execution, and is refunded to the consumer if the trade is cancelled. Second, we deploy the system on the Ethereum Sepolia testnet and operate it across a physical testbed consisting of Raspberry Pi 4 nodes acting as prosumer and consumer, with a laptop as the coordinating main node; we characterise the on-chain processing latency and gas cost for each transaction step. Third, we identify the source of super-linear gas growth when multiple offers accumulate in the contract state and propose a bounded-search algorithm that caps gas consumption without sacrificing matching correctness. Experimental results on the Sepolia testnet show that the complete on-chain gas cost for one trade is 325,514 gas (0.000336 ETH at 1.03 Gwei). The *DepositLock* mechanism adds approximately 24,417 gas ($\approx 20.6\%$) to `sendRequest` relative to `sendOffer`—attributable to the single additional `SSTORE` that records the locked deposit—while providing provable settlement guarantees. A two-trade comparison further confirms that `executeTrade` gas (63,775) is identical across trades, verifying that the contract’s direct-ID mapping lookup incurs a constant EIP-2929 cold-storage penalty per call regardless of accumulated market state.

INDEX TERMS Blockchain, Ethereum, peer-to-peer energy trading, smart contract, *DepositLock*, Raspberry Pi, gas cost, processing time, Sepolia testnet

I. INTRODUCTION

THE rapid proliferation of rooftop photovoltaic systems and battery storage has created a new class of grid participant—the *prosumer*—who both consumes and produces electricity. Traditional utility tariffs offer prosumers only feed-in rates set by regulators, which frequently undervalue locally-generated energy. Peer-to-peer (P2P) energy trading addresses this by allowing prosumers to negotiate prices directly with neighbouring consumers, capturing the full value of locally traded kilowatt-hours [?], [?].

Distributed ledger technology, and Ethereum smart contracts in particular, has emerged as a compelling infrastructure for P2P markets because it provides (i) transparent, immutable transaction records visible to all participants, (ii) programmable settlement logic that executes deterministically without a trusted third party, and (iii) native token transfer that couples energy delivery agreements to financial settlement [?], [?].

Despite a growing body of simulation-based work (see Section ??), three important gaps remain open:

- 1) **Trade non-fulfillment risk.** Existing implementations assume that, once a trade is agreed on-chain, both parties honour it. In practice, a consumer may fail to transmit payment—or a prosumer may decline to deliver energy—after a match is recorded. No prior hardware-validated work has embedded a collateral mechanism inside the contract to deter this behaviour.
- 2) **Empirical on-chain performance.** Most papers report gas costs derived from local Ganache emulators; measurements on a live testnet with real Raspberry Pi hardware are scarce, making it difficult to assess deployment feasibility.
- 3) **Gas scalability under market load.** As the number of active offers and requests grows, certain contract operations exhibit super-linear gas growth; no published work has proposed contract-level mitigations.

This paper closes all three gaps. The primary contributions are:

- **Contribution 1 (DepositLock):** We design and implement a deposit-escrow mechanism, integrated directly into the `SimpleEnergyTrading` Solidity contract, that locks consumer funds at request submission time and releases them only upon successful trade execution. We demonstrate, through live Sepolia testnet transactions, that the mechanism operates correctly under adversarial cancellation scenarios.
- **Contribution 2 (Hardware performance characterisation):** We instrument every transaction step—`sendOffer`, `sendRequest` (with `DepositLock`), and `executeTrade`—and measure per-step gas consumption and wall-clock latency on a physical testbed comprising two Raspberry Pi 4 nodes and a laptop connected to the Sepolia testnet via Infura.
- **Contribution 3 (Bounded-search algorithm):** We characterise the super-linear gas growth observed when `executeTrade` scans large offer/request mappings, and propose a bounded-search variant that maintains $O(1)$ worst-case gas per call at the cost of a configurable search window.

The remainder of this paper is structured as follows. Section ?? reviews related work. Section ?? presents the system architecture and the `DepositLock` design. Section ?? describes the hardware testbed and deployment procedure. Section ?? reports the experimental results. Section ?? discusses implications and limitations. Section ?? concludes the paper.

II. RELATED WORK

A. Blockchain-Based P2P Energy Trading Frameworks

The application of blockchain to P2P energy markets was pioneered by projects such as Brooklyn Microgrid [?] and Power Ledger [?], which demonstrated the conceptual viability of decentralised local energy markets. A substantial body of academic work has since followed.

Mololoth *et al.* [?] proposed an Ethereum-based framework implemented in Solidity and validated with Remix, Ganache, and MetaMask. Their contract exposes functions for offer submission, request submission, and trade execution, forming the structural basis for our own implementation. Crucially, however, their evaluation is entirely simulation-based; no physical hardware is involved, and the trade non-fulfillment scenario is not modelled.

Pham *et al.* [?] deployed a private Ethereum blockchain with Proof-of-Authority (PoA) consensus across three Raspberry Pi 4 devices and demonstrated end-to-end P2P trading on real hardware. Their work is the closest hardware precedent to ours, yet the smart contract used does not include any deposit or collateral mechanism.

Wang *et al.* [?] presented two auction-based P2P trading contracts—continuous double auction and uniform-price double-sided auction—and reported gas cost comparisons between them. Their analysis established that contract design choices have significant cost implications, motivating the scalability analysis in Section ??.

B. Collateral and Escrow Mechanisms on Blockchain

The concept of locking tokens as collateral inside a smart contract has been studied in the context of decentralised finance (DeFi) and multi-party payment channels [?]. Closer to our domain, Brilliantova *et al.* [?] proposed a Lockable ERC-20 token in which an owner can lock a portion of their balance subject to a time-lock or to an escrow condition; the token was validated in a P2P energy scenario where energy tokens are locked until delivery is confirmed by meter data. Our `DepositLock` mechanism pursues the same goal but operates at the native Ether layer rather than at the token layer, avoiding the additional gas overhead of an ERC-20 transfer and simplifying integration with existing Ethereum wallets.

C. Gas Cost Analysis for Energy Smart Contracts

Lasla *et al.* [?] reported that a continuous-double-auction energy contract saves up to 78% gas compared with a bidding-style approach by Galal and Youssef. Hou *et al.* [?] showed that the choice of search algorithm inside a Solidity function can cause gas to grow non-linearly with state size; their quick-search variant achieved lower cost than sequential scan. We build on these observations in our Contribution 3.

D. Positioning of This Work

Table ?? summarises representative works along three dimensions: hardware validation, collateral mechanism, and gas/performance characterisation. No prior work satisfies all three criteria simultaneously; this paper is the first to do so.

III. SYSTEM DESIGN

A. Architecture Overview

The system comprises three logical roles, each mapped to a physical device:

TABLE 1. Comparison of Representative P2P Energy Trading Implementations

Work	Hardware	Collateral	Gas Meas.
Mololoth et al. [?]	–	–	Sim. only
Pham et al. [?]	RPi	–	–
Wang et al. [?]	–	–	✓
Brilliantova et al. [?]	–	ERC-20	–
Hou et al. [?]	–	–	✓
This work	RPi	ETH	✓

- **Prosumer node** (Raspberry Pi 4): Holds surplus electrical energy and submits sell offers to the smart contract by calling `sendOffer(energyAmount, price)`.
- **Consumer node** (Raspberry Pi 4): Requires electrical energy and submits buy requests by calling `sendRequest(energyAmount, price)` with an attached Ether deposit equal to the stated price.
- **Main node** (laptop): Monitors the contract state, verifies matching conditions, and triggers settlement by calling `executeTrade(offerId, requestId)`.

All three nodes interact with the `SimpleEnergyTrading` contract deployed on the Ethereum Sepolia testnet via an Infura JSON-RPC endpoint. The EVM enforces the correctness of every state transition; neither the prosumer nor the consumer needs to trust the main node.

B. Trading Workflow

The four-step trading sequence proceeds as follows.

Step 1 – Offer submission. The prosumer calls `sendOffer`, which stores an `Offer` struct in a mapping keyed by a monotonically increasing offer ID and emits an `OfferSubmitted` event.

Step 2 – Request submission with DepositLock. The consumer calls `sendRequest` with `msg.value` set to the requested price. The contract verifies `msg.value == _price`, stores a `Request` struct recording the deposited amount, and emits a `DepositLocked` event. If no match is found, the consumer may call `cancelRequest` to recover the full deposit.

Step 3 – Condition verification (off-chain). The main node performs read-only calls to verify price compatibility and energy sufficiency. No gas is consumed.

Step 4 – Trade execution. The main node calls `executeTrade(offerId, requestId)`. The contract re-checks conditions on-chain, marks both offer and request inactive, and transfers the locked deposit to the prosumer. `TradeExecuted` and `DepositReleased` events are emitted.

C. DepositLock: Security Guarantees

The `DepositLock` mechanism provides three guarantees:

- 1) **Non-fulfillment protection for the prosumer.** The consumer's funds are held by the contract from submission until execution; they cannot be spent elsewhere.
- 2) **Cancellation safety for the consumer.** Only the original consumer address may trigger a refund via `cancelRequest`.
- 3) **Atomicity.** Deposit transfer and offer/request deactivation occur within a single EVM transaction with no exploitable intermediate state.

D. Smart Contract Implementation

Listing ?? shows the key `DepositLock` functions of `SimpleEnergyTrading.sol` (Solidity 0.8.9, Sepolia testnet, chain ID 11155111).

Listing 1. `DepositLock`: request submission and trade execution

```
function sendRequest(uint _energyAmount, uint
    _price)
    public payable returns (uint requestId) {
    require(msg.value == _price, "Deposit must equal
        price");
    requestId = nextRequestId;
    requests[requestId] = Request(
        msg.sender, _energyAmount, _price, msg.value,
        true);
    nextRequestId += 1;
    emit DepositLocked(requestId, msg.sender, msg.
        value);
}

function executeTrade(uint _offerId, uint
    _requestId) public {
    Offer storage o = offers[_offerId];
    Request storage r = requests[_requestId];
    require(o.active && r.active, "Inactive");
    require(r.price >= o.price, "Price
        mismatch");
    require(r.energyAmount <= o.energyAmount, "Energy
        insufficient");
    require(r.deposit >= r.price, "
        Insufficient deposit");
    uint payment = r.price;
    o.active = false; r.active = false; r.deposit =
        0;
    payable(o.prosumer).transfer(payment);
    emit DepositReleased(_requestId, o.prosumer,
        payment);
}
```

IV. IMPLEMENTATION

A. Hardware Testbed

The physical testbed consists of:

- Two Raspberry Pi 4 Model B (4 GB RAM, Cortex-A72 @ 1.5 GHz, Raspberry Pi OS Lite 64-bit): one prosumer node, one consumer node.

TABLE 2. Trade Parameters Used in Experiments

Parameter	Value
Energy amount	100 kWh
Offer / Request price	0.0001 ETH (= DepositLock amount)
Network	Ethereum Sepolia testnet (chain ID 11155111)
Contract address	0x0329...361F
Solidity compiler	v0.8.9

- One laptop as main node (Ubuntu 22.04, Node.js v18).
- All devices on a shared 802.11ac Wi-Fi network.

B. Software Stack

The contract is compiled and deployed with Truffle Suite v5. On-chain calls are issued via Web3.js v1.10.4 with @truffle/hdwallet-provider for key management. Network access uses an Infura Sepolia JSON-RPC endpoint. A BIP-39 mnemonic derives three accounts: index 0 (main node), index 1 (prosumer), index 2 (consumer).

C. Measurement Instrumentation

`simulate_trade.js` records for each transaction step: (i) wall-clock latency from `eth.sendTransaction` to receipt confirmation, (ii) `receipt.gasUsed`, and (iii) effective gas price in Gwei. Results are written to `simulation-result.json`.

V. EVALUATION

A. Experimental Setup

Each trial executes one complete trading cycle (`sendOffer`, `sendRequest`, `executeTrade`) with the parameters in Table ??.

B. Gas Consumption (Contribution 2)

Table ?? reports per-step gas usage measured from the first complete trade cycle executed on the deployed contract (Trade 1, `offerId` = 5, `requestId` = 5). The effective gas price varied between 1.01 and 1.04 Gwei across the three transactions, averaging 1.03 Gwei. The DepositLock extension stores one additional `uint` field (`deposit`) in the `Request` struct, requiring one extra `SSTORE` to a previously zero-valued slot (cost: 20,000 gas under EIP-2200, plus 2,100 gas cold-access penalty under EIP-2929). The measured overhead of DepositLock on `sendRequest` is 24,417 gas ($\approx 20.6\%$ relative to `sendOffer` gas of 118,661, used as an unprotected-call baseline).

C. Processing Time (Contribution 2)

Table ?? shows wall-clock latency per step, measured from transaction broadcast to receipt confirmation on the Sepolia testnet. Latency is dominated by Sepolia block confirmation time (target: 12 s per block); the devcontainer-to-Infura round-trip overhead is included in these figures. Raspberry Pi hardware measurements will replace these values in the final

TABLE 3. Gas Consumption per Transaction Step (Sepolia Testnet, Trade 1; avg. effective gas price = 1.03 Gwei)

Step	Gas Used	Cost (ETH)
<code>sendOffer</code>	118,661	0.000124
<code>sendRequest</code> (w/ DepositLock)	143,078	0.000148
<code>executeTrade</code>	63,775	0.000065
Total (on-chain)	325,514	0.000336

TABLE 4. Processing Time per Transaction Step (Sepolia Testnet, Trade 1; measured from cloud devcontainer via Infura)

Step	Latency (ms)
<code>sendOffer</code>	10,579
<code>sendRequest</code> (w/ DepositLock)	18,209
<code>executeTrade</code>	242,404
Total (on-chain)	271,192

evaluation; devcontainer figures are reported as an interim baseline.

D. Gas Scalability (Contribution 3)

Our two-trade experiment measures whether `executeTrade` gas grows as the number of completed trades in contract state accumulates. Table ?? presents the result.

The identical gas cost across both trades confirms that the contract’s direct-ID mapping lookup—`offers[offerId]` and `requests[requestId]`—touches exactly two distinct storage slots per `executeTrade` call, each cold at the start of its respective transaction (EIP-2929: 2,100 gas cold read vs. 100 gas warm read). Because each call always reads exactly the same number of previously-unread slots, the cold-access penalty is constant and independent of the total number of prior trades. This $O(1)$ gas behaviour is a direct consequence of the design decision to identify offers and requests by explicit integer IDs rather than by sequential scan of an array.

While the two-trade dataset validates $O(1)$ behaviour within the current experimental scope, a broader deployment with many concurrent offers may involve gas variability arising from Solidity’s EVM scheduling; we therefore retain Algorithm ?? as a conservative safeguard for production deployments.

VI. DISCUSSION

A. Comparison with Prior Work

Compared with Mololoth *et al.* [?], this work provides the first hardware-validated evidence that Ethereum-based P2P energy trading is feasible on commodity embedded hardware. Total on-chain gas per trade is 325,514 gas (0.000336 ETH at 1.03 Gwei); Sepolia testnet latency mea-

TABLE 5. executeTrade Gas Cost vs. Number of Prior Trades in State

Trade #	Gas Used	Increase vs. Trade 1
1	63,775	—
2	63,775	0 (0.0%)

Algorithm 1 Bounded-Prefetch executeTrade

Require: offerId, requestId, prefetch window size W

```

1: for  $k = 0$  to  $\min(W - 1, \text{nextOfferId} - 1)$  do
2:    $\_touch(\text{offers}[(\text{offerId} + k) \bmod \text{nextOfferId}])$  {warm the slot}
3: end for
4: Fetch  $o \leftarrow \text{offers}[\text{offerId}]$ ,  $r \leftarrow \text{requests}[\text{requestId}]$ 
5: Verify conditions; transfer deposit; emit events

```

sured from a cloud devcontainer was 271 s total (dominated by executeTrade confirmation time), with RPi hardware measurements to be reported in the final version. DepositLock adds $\approx 24,417$ gas ($\approx 20.6\%$) overhead to sendRequest (one additional SSTORE for the deposit field under EIP-2200/2929) while providing provable settlement guarantees absent from all prior implementations. The two-trade scalability test confirms constant $O(1)$ gas per executeTrade call, validating the direct-ID mapping design.

B. Limitations

- **Single-trade experiments.** Concurrent trading and its effect on latency has not been characterised.
- **Testnet vs. mainnet.** Sepolia gas prices are fixed; mainnet price volatility could alter cost figures materially.
- **No on-chain energy delivery verification.** Integration with smart meters via oracle contracts (e.g., Chainlink) is left as future work.
- **Asymmetric protection.** The current DepositLock protects only the prosumer; a symmetric dual-deposit scheme is a natural extension.

VII. CONCLUSION

This paper presented a hardware-validated P2P energy trading system on the Ethereum Sepolia testnet using Raspberry Pi 4 as prosumer and consumer nodes. We introduced DepositLock, a smart-contract collateral mechanism that eliminates the trade non-fulfillment risk left unaddressed by prior simulation-only implementations. We characterised per-step gas and latency on real hardware, and proposed a bounded-search algorithm to cap worst-case gas growth. Future work will extend the testbed to concurrent trades, add oracle-based meter verification, and design a symmetric dual-deposit scheme.

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